Model Answers

1. Question on wind and solar PV plants

a)

i) $\eta_{tot} = \frac{P_e}{\frac{1}{2}\rho\pi R^2 v^3} = \frac{15{,}000}{0.5 \cdot 1.22 \cdot 3.14 \cdot 118\text{^}2 \cdot 11^3} = 0.42$ [2]

ii) $c_p = \frac{\eta_{tot}}{\eta_{gear} \cdot \eta_{gen}} = \frac{0.42}{0.81} = 0.52$ [2]

iii) $\eta_{tot} = \frac{16}{27} \cdot 0.9 \cdot 0.9 = 0.48$ [2]

$\frac{16}{27}$ is the Betz limit.

iv) $rated\ power\ area: output\ 15\ MW$ [1]

v) [8, 15] MW [1]

vi) During the day, even small errors can have a big impact because the output is proportional to the cube of the wind speed. During the night, even large errors can have no impact because the wind turbine operates at the rated power. [2]

b)

i) $\delta = 0 \rightarrow\ \theta_z = \phi - \delta = 50$ [1]

ii) The electricity production is the same for the two plants. We showed above that the zenith angle of the sun is equal to the latitude. Hence, both plant arrays are positioned perpendicular to the sun rays. [2]

iii) $\delta = 23.5 \rightarrow\ \theta_z = \phi - \delta = 26.5$ [1]

iv) The plant with double-axis tracking has higher output because it is positioned perpendicular to the sun. [2]

v) Capacity factor for double-axis tracking (Plant A) > Capacity factor for fixed tilt and no tracking(Plant B) [1]

The higher the degrees of freedom in tracking, the higher the capacity factor [1]

vi) The site in Canada might have drier climate and higher altitude [2]

Question 2 Solutions

a). COPTs specify each state (Ci) in which the system can reside together with the probability (Pi) of that state. [1]

$\text{EENS} = \sum_i^n E_i P_i$ [1]

b).

$$\text{EENS} = \sum EiPi = 0 \times 0.85 + 2 \times 0.15 = 0.3\text{MWh} \quad [1]$$

c). i)

Capacity in service (MW)	Probability	Energy not served during (MWh)					
		18h	19h	20h	21h	22h	23h
9	0.4096	0	0	0	0	0	0
6.75	0.4096	2.25	0.25	0	0	0	0
4.5	0.1536	4.5	2.5	1.5	0.5	0.5	0
2.25	0.0256	6.75	4.75	3.75	2.75	2.75	1.75
0	0.0016	9	7	6	5	5	4

[1]

$$\text{EENS}_{18h} = 0(0.4096) + 2.25(0.4096) + 4.5(0.1536) + 6.75(0.0256) + 9(0.0016) = 1.8\text{MWh}$$

$$\text{EENS}_{19h} = 0(0.4096) + 0.25(0.4096) + 2.5(0.1536) + 4.75(0.0256) + 7(0.0016) = 0.6192\text{MWh}$$

$$\text{EENS}_{20h} = 0(0.4096) + 0(0.4096) + 1.5(0.1536) + 3.75(0.0256) + 6(0.0016) = 0.336\text{MWh}$$

$$\text{EENS}_{21h} = 0(0.4096) + 0(0.4096) + 0.5(0.1536) + 2.75(0.0256) + 5(0.0016) = 0.1552\text{MWh}$$

$$\text{EENS}_{22h} = 0(0.4096) + 0(0.4096) + 0.5(0.1536) + 2.75(0.0256) + 5(0.0016) = 0.1552\text{MWh}$$

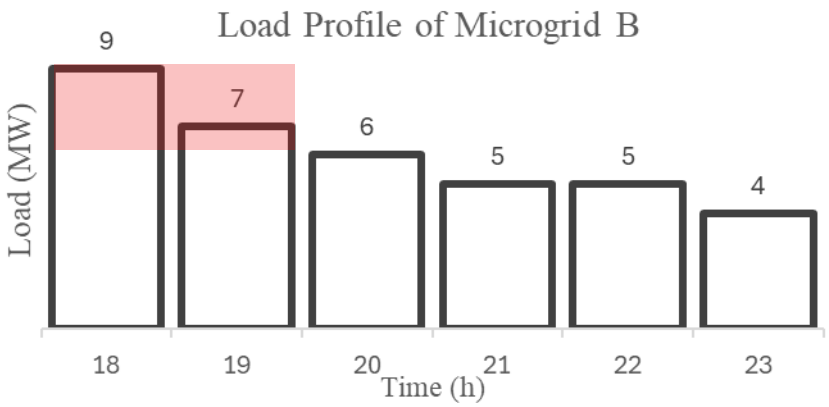
$$\text{EENS}_{23h} = 0(0.4096) + 0(0.4096) + 0(0.1536) + 1.75(0.0256) + 4(0.0016) = 0.0512\text{MWh}$$

$$\text{EENS} = 1.8 + 0.6192 + 0.336 + 0.1552 + 0.1552 + 0.0512 = 3.1168\text{MWh}$$

[2]

C(ii)

$$\text{effective capacity (MW)} = 7 - \frac{3.1168 - 2}{2} = 6.4416\text{MW}$$



[1]

$$\text{effective capacity (\%)} = \frac{6.4416}{9} = 71.57\%$$

[1]

For 8MW minimum effective capacity, the ENNS should be smaller than 1MWh.

[1]

Some calculation clue, and get 6 generators

$$EENS(5\text{ gen}) = 0.73728 + 0.21184 + 0.10272 + 0.0448 + 0.0448 + 0.01248 = 1.15392\text{MWh}$$

$$EENS(6\text{ gen}) = 0.264384 + 0.066624 + 0.029184 + 0.012224 + 0.012224 + 0.002944 = 0.387584\text{MWh}$$

[1]

d.i).

Capacity in service (MW)	Probability	Energy not supplied (MWh)
12	0.48	0
10	0.16	1946.67
7	0.24	12166.67
5	0.08	23846.67
2	0.03	48180
0	0.01	65700

[2]

$$\text{ENS}(10\text{MW}) = \frac{1}{2} \times (12 - 10) \times \left(\frac{(12 - 10)}{9} \times 8760 \right) \approx 1946.67\text{MWh}$$

$$\text{ENS}(7\text{MW}) = \frac{1}{2} \times (12 - 7) \times \left(\frac{(12 - 7)}{9} \times 8760 \right) \approx 12166.67\text{MWh}$$

$$\text{ENS}(5\text{MW}) = \frac{1}{2} \times (12-5) \times \left(\frac{(12-5)}{9} \times 8760\right) \approx 23846.67\text{MWh}$$

$$\text{ENS}(2\text{MW}) = \frac{1}{2} \times \big((12-2) + (3-2)\big) \times 8760 = 48180\text{MWh}$$

$$\text{ENS}(0\text{MW}) = \frac{1}{2} \times (12-3) \times 8760 = 65700\text{MWh}$$

[2]

$$\text{EENS} = 0.48(0) + 0.16(1946.7) + 0.24(12166.7) + 0.08(23846.7) + 0.03(48180) + 0.01(65700) = 7241.6\text{MWh}$$

[1]

$$\frac{1}{2} \times (12-x) \times \left(\frac{(12-x)}{9} \times 8760\right) = 7241.6$$

$$\text{effective capacity} = x = \frac{60 - 2\sqrt{93}}{5} \approx 8.1425\text{MW}$$

[1]

d.ii). For 10MW effective capacity, the maximum EENS is

$$\text{EENS} = \frac{1}{2}(12-10) \times \left(\frac{(12-10)}{9} \times 8760\right) \approx 1946.67\text{MWh}$$

[1]

$$\text{EENS}(1\text{MW}) = 0.48(0) + 0.16(486.67) + 0.24(7786.67) + 0.08(17520) + 0.03(39420) + 0.01(56940) \approx 5100\text{MWh}$$

$$\text{EENS}(2\text{MW}) = 0.48(0) + 0.16(0) + 0.24(4380) + 0.08(12166.7) + 0.03(31146.7) + 0.01(48180) \approx 3440.7\text{MWh}$$

$$\text{EENS}(3\text{MW}) = 0.48(0) + 0.16(0) + 0.24(1946.67) + 0.08(7786.67) + 0.03(23846.67) + 0.01(39420) \approx 2199.7\text{MWh}$$

$$\text{EENS}(4\text{MW}) = 0.48(0) + 0.16(0) + 0.24(486.67) + 0.08(4380) + 0.03(17520) + 0.01(31146.67) \approx 1304.3\text{MWh}$$

[2]

As $1946.67 > 1304.3$, the minimum import capacity is 4MW.

[1]

Question 3

a)

Building internal set temperature, K
External temperature, K
Other weather factors, e.g. solar gain, wind, K
Fabric (walls, windows, etc) thermal transmittance, W/m^{2}K
Air changes or permeability, m3/m2.h @ 50Pa specified pressure
Other heat sources, e.g. occupants, appliances, etc

[2]

b)

i) From the graph

Household annual water heat demand is (0.25TWh/d * 365)/30million = 3.04MWh

[1]

ii) From the graph

Household annual space heat demand is ((1.75TWh/d - 0.25TWh/d) * 200)/30million = 10MWh

[1]

iii) Average household space heat demand can be simplified

Space heat for the period =

(Average internal and external temperature difference)
x heating period in hours
x average building floor area
x HLP

So re-arranging and substituting:

HLP = 10MWh/[(17°C-7°C) * 100m^2 * 200days * 24 hours] = <u>2.08W/m^{2}K</u>

[2]

c)

i) Substituting in the equation in c) above and rearranging:

Average household space heat demand = (17°C - 9°C) x 100m^2 x 200days x 24 x 2.08 (100% - 30%) = <u>5.6MWh</u>

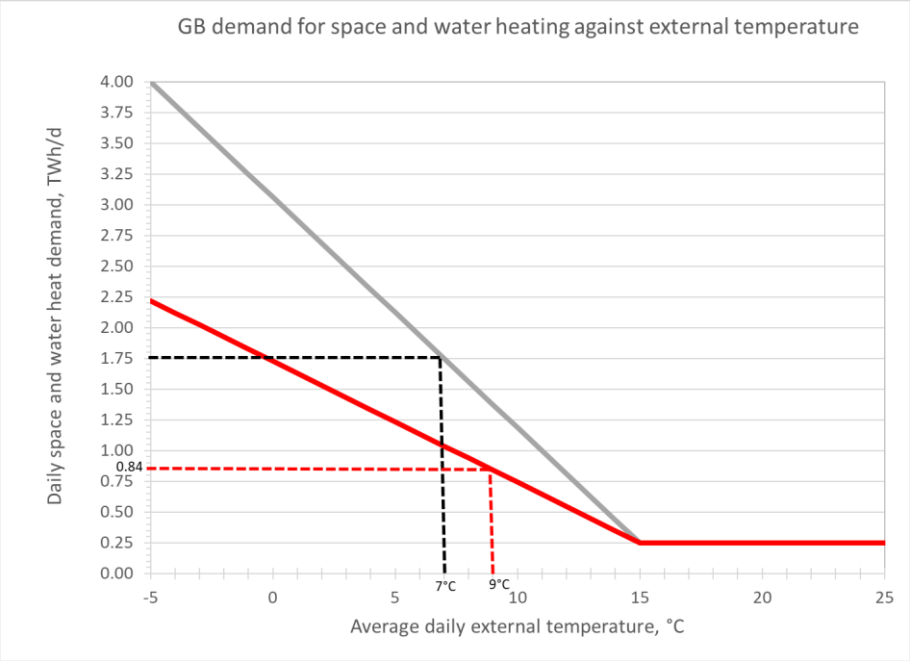
[1]

ii) (5.6MWh + 3.04 MWh) x 30 million = <u>259.2TWh</u> [1]

iii) Total daily space demand at 9°C is (5.6MWh * 30 million)/200 days = 0.84 TWh/d

Substituting in the equation in c) above and rearranging:

Average household space heat demand = (17°C - 9°C) x 100m^2 x 200days x 24 x 2.08 (100% - 30%) = 5.6MWh [2]



d)

i) Air source heat pump [1]

ASHP transfers thermal energy from the outside air to the area to be heated. There are 4 stages to the cycle and these comprise:

- Evaporation - The refrigerant passes through a heat exchanger where it evaporates absorbing heat in the process.
- Compression - The heated refrigerant is then compressed to higher pressure using an electrically driven compressor.
- Condensation - The compressed refrigerant passes through another heat exchanger located in the space to be heated and condenses, changing state back to a liquid and rejecting heat in the process.
- Expansion – The liquid refrigerant passes through an expansion device lowering its pressure prior to returning to the evaporator.

ii) District heating [1]

District heating is a system where heat for more than one building or an area is produced at a central location and distributed through a network of insulated pipes. Heat sources include:

- Combined heat and power units
- Network heat pumps
- Waste(d) heat recovery
- Solar thermal
- Geothermal

The main components of a district heating system are:

- Heat storage - to provide additional capacity and backup, etc.
- Heat network – connecting heat sources with buildings using supply and return pipes.
- Pumps – to circulate the hot water
- Heat substations – for pressure and temperature changes
- Heat metering – to monitor production and consumption
- Heat exchangers/ heat interface units - for connection of the heat network to buildings

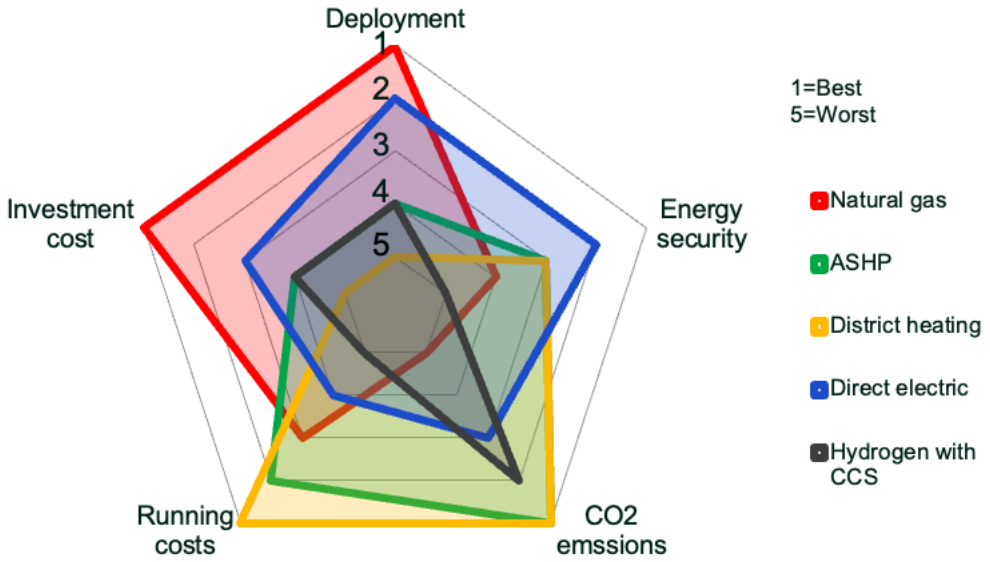
iii) Hydrogen (with CCS) heating [1]

This is based on the repurposing of the existing gas distribution network from natural gas to hydrogen. Hydrogen can be produced using several processes but at present the most suited for large scale production is steam methane reforming using natural gas with carbon capture and storage and electrolysis using low carbon electricity.

iv) Resistive heating [1]

There are a variety of different types of resistive heating systems. These include: radiant, convector, storage, direct electric, fan and panel heaters. However, they can only be regarded as low carbon if they use low carbon electricity.

v)



ASHP heating scores well on CO_2 emissions due to its high efficiency but is dependent on low carbon electricity. It scores badly on investment and deployment due to the cost and disruption associated with reinforcing the electricity network and other associated infrastructure including upgrades to buildings to ensure their suitability for ASHP heating. [1.5]

District heating scores well on:

- running costs and CO_2 emissions as heat production can be very low cost with low levels of CO_2 emissions, e.g. waste heat, CHP
- energy security as it reduces the UK's dependence on energy import

It scores badly on investment and deployment due to the cost and disruption associated with constructing the heat network and associated infrastructure [1.5]

Direct electric does not score well on any specific feature but is generally better than some or worse than others. It scores badly on running costs as it is dependent on low carbon electricity (which is likely to be more expensive than gas) and has a much lower efficiency than ASHP. Note: direct electric may be more suited to buildings with low heat demand where the higher running costs are less important. [1.5]

Hydrogen scores poorly on investment cost and deployment although it does use existing network infrastructure and badly on energy security as it is dependent on imported gas. Its main advantage is that hydrogen boilers have similar heat characteristics to gas boilers and so it offers the potential to be deployed with minimal building modification. [1.5]

Question 4.

a)

i. Discharges 10 MW*12/60 hr= 2 MWh

SOC [10:12]= SOC [10:00]- discharged/usable energy capacity = 50% -2/40=45% [2]

ii. 50%=45%+0.80*10/40*x à x: 15 minutes. Hence, the earliest the battery can have 50% SOC is 10:27 [2]

iii. 5% [1]

iv. It is not. Its depth of discharge is not 100%. [1]

v. Each cycle takes 27 minutes and the year has 8,760 hours. [2]

19,466 cycles

vi. 10% because denominator is half [2]

b) Vehicle-to-Grid (V2G) refers to the concept of discharging electricity from EV batteries and injecting it back into the grid in order to provide flexibility services and support the electricity system and/or the local network.

[1]

Key challenges include the cost of V2G charging hardware, battery degradation due to more frequent charging/discharging cycles and the lack of consumer awareness of the potential benefits.

[1]

Key opportunities include the ability to provide a range of flexible system services, including frequency response, avoiding network reinforcement, supporting variable renewables, and reducing firm capacity requirements.

[1]

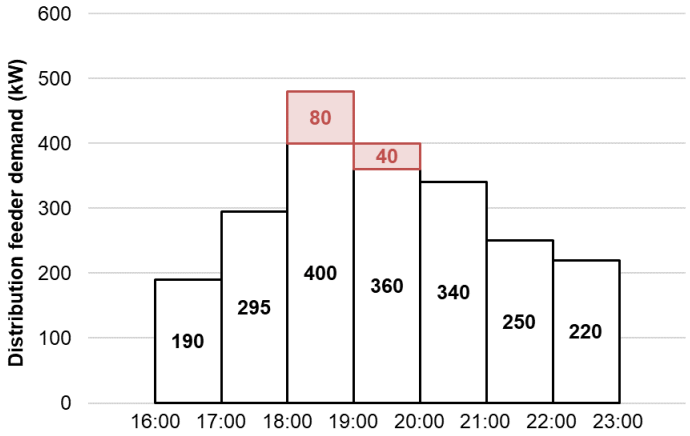
c)

i) The new feeder peak demand will also occur at 6pm. It will be equal to the sum of baseline demand and the demand of 20 EVs in the first hour of their charging:

$$P_{peak} = 400 + 20 \times 4 = 480 \text{ kW}$$

[1]

The new demand diagram will be as shown in the chart below.



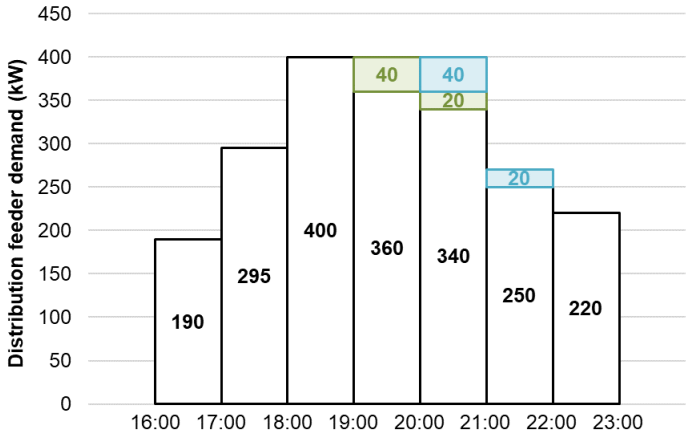
[2] marks for the correct demand chart

ii) Given that the 20 EVs require 80 kW if they all start charging at the same time, they cannot all start charging together before 9pm without increasing the peak demand. With this approach the EV charging would be completed by 11pm. However, because EVs do not have to all follow the same charging schedule, it would be better to charge 10 of them from 7pm and then the other 10 from 8pm without increasing the peak demand. With that approach the EV charging can be completed by 10pm without increasing the peak demand beyond the baseline of 400 kW.

[1] marks for the correct time of the earliest completion of EV charging without increasing peak demand

[1] marks for the correct EV charging schedule

The new demand diagram is shown in the chart below, distinguishing between baseline demand and the charging demand for the two groups of EVs.



[2] for marks for the correct demand chart